



Lecture 1.2: Representation of Power System

EEE 321: Power System I

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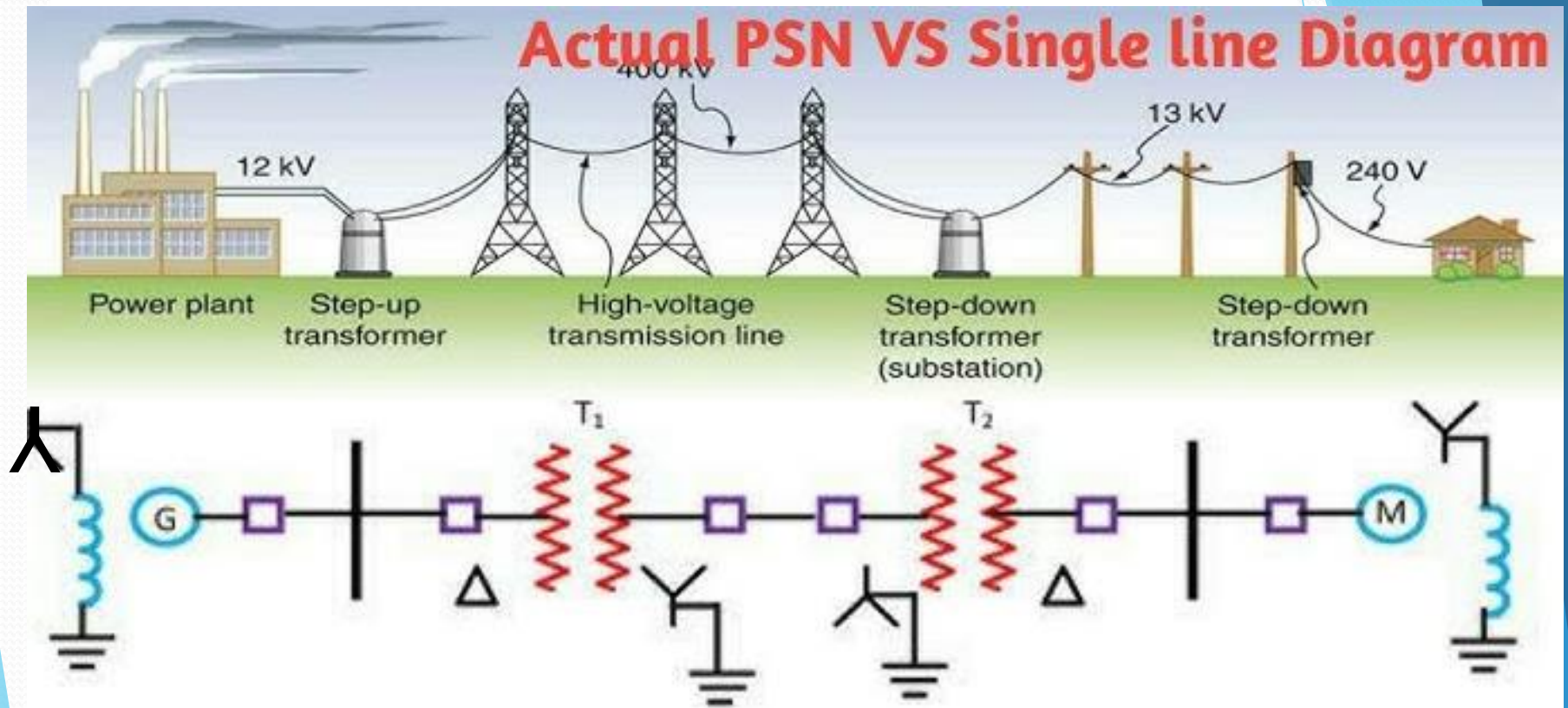
Presentation Outline

Representation of Power System

- ❑ Single Line Diagram
- ❑ Impedance Diagram
- ❑ Reactance Diagram

Representation of Power System

SINGLE LINE OR ONE LINE DIAGRAM



Representation of Power System

IMPEDANCE DIAGRAM

In order to calculate the performance of a power system under load condition or upon the occurrence of a fault, the one line diagram is used to draw the single-phase or per phase equivalent circuit of the system.

Impedance and reactance diagrams are the simplified equivalent circuits of single line or one line diagrams of power systems.

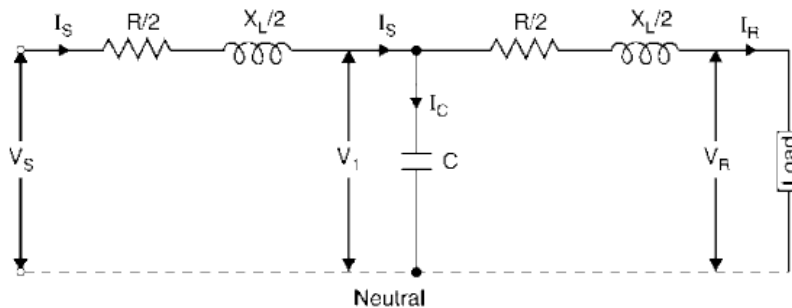
- ❑ A further simplification from the single line diagram is to draw the diagram with impedances only.
- ❑ In impedance diagram, each component is represented by its approximate or simplified equivalent circuit.
- ❑ The impedance diagram is drawn on single-phase basis for use under balanced 3 phase system.
- ❑ The impedance diagram does not include the current limiting impedances shown in the one-line diagram because no current flows in the ground under balanced condition.

Representation of Power System

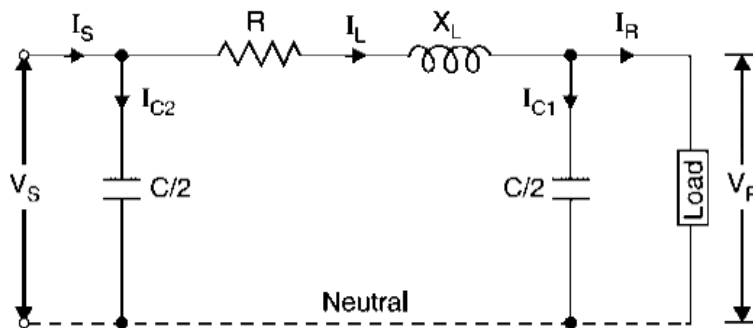
IMPEDANCE DIAGRAM

Per phase equivalent circuit of power system components

Transmission Line

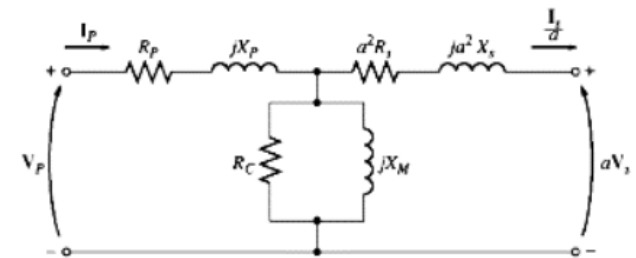


(i) Nominal T method

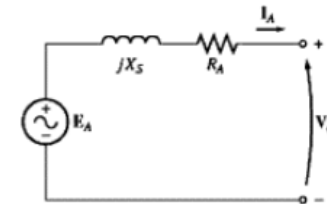


(ii) Nominal π method

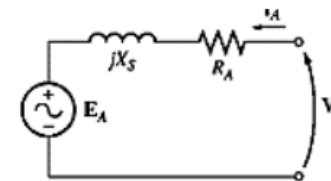
Transformer



Synchronous Generator



Synchronous Motor



Representation of Power System

IMPEDANCE DIAGRAM

Per phase equivalent circuit of power system components

Transmission line: The equivalent circuit of the transmission line is represented with sufficient accuracy by the nominal π model having the total resistance and inductive reactance of the line in its series arm and the total capacitance to neutral divided between its shunt arms.

Transformer: Resistance, leakage reactance, and path for magnetizing current are shown for each transformer.

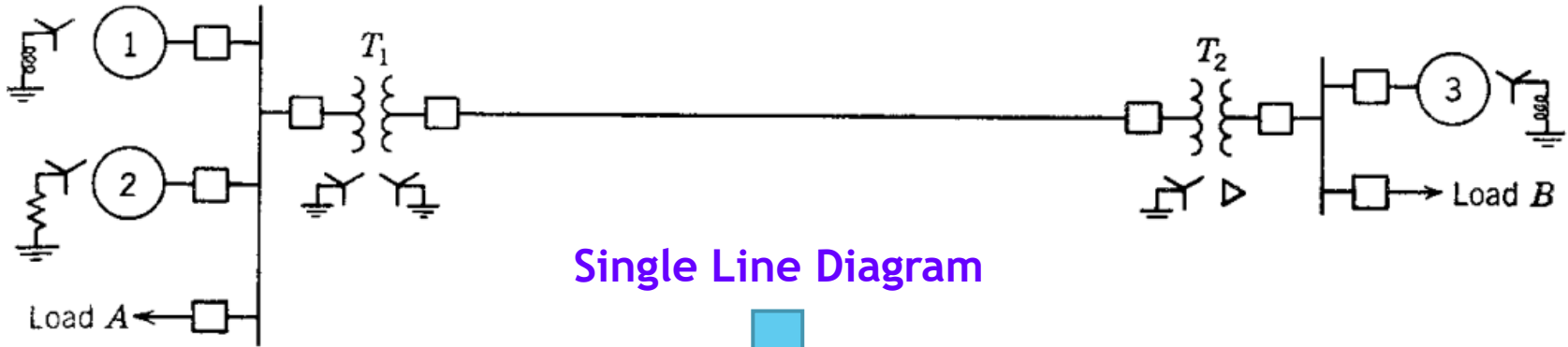
Generator: A generated voltage in series with appropriate values of resistance and reactance represents each generator.

Load: If a load study is to be made, the lagging loads A and B are represented by resistance and inductive reactance in series.

Note: The impedance diagram does not include the current-limiting impedances shown in the one-line diagram between the neutrals of the generators and ground because no current flows in the ground under balanced conditions and the neutrals of the generators are at the potential of the neutral of the system.

Representation of Power System

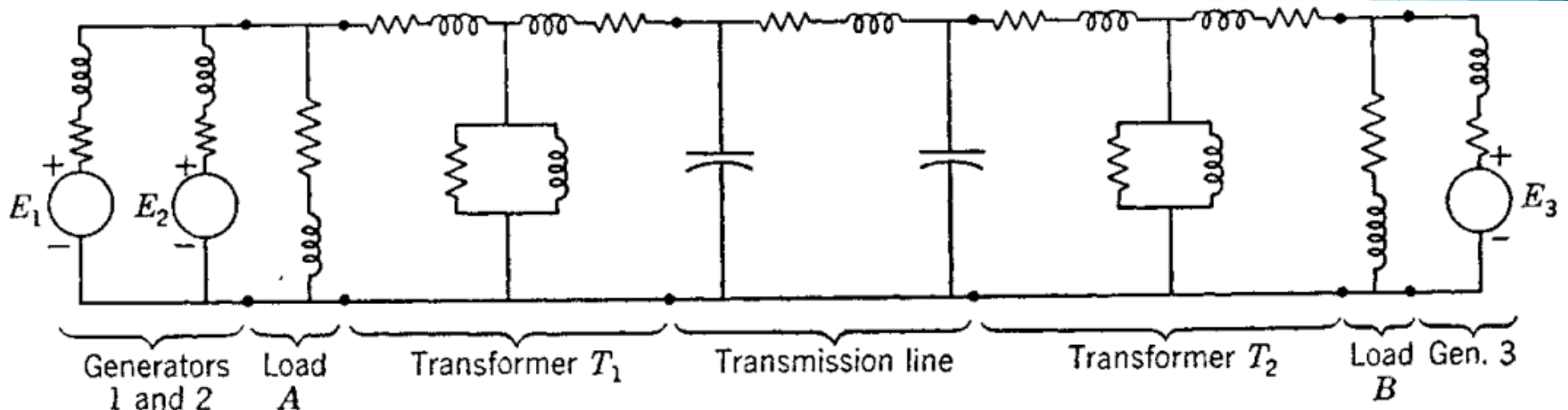
SINGLE LINE DIAGRAM TO IMPEDANCE DIAGRAM



Single Line Diagram



Impedance Diagram



Representation of Power System

REACTANCE DIAGRAM

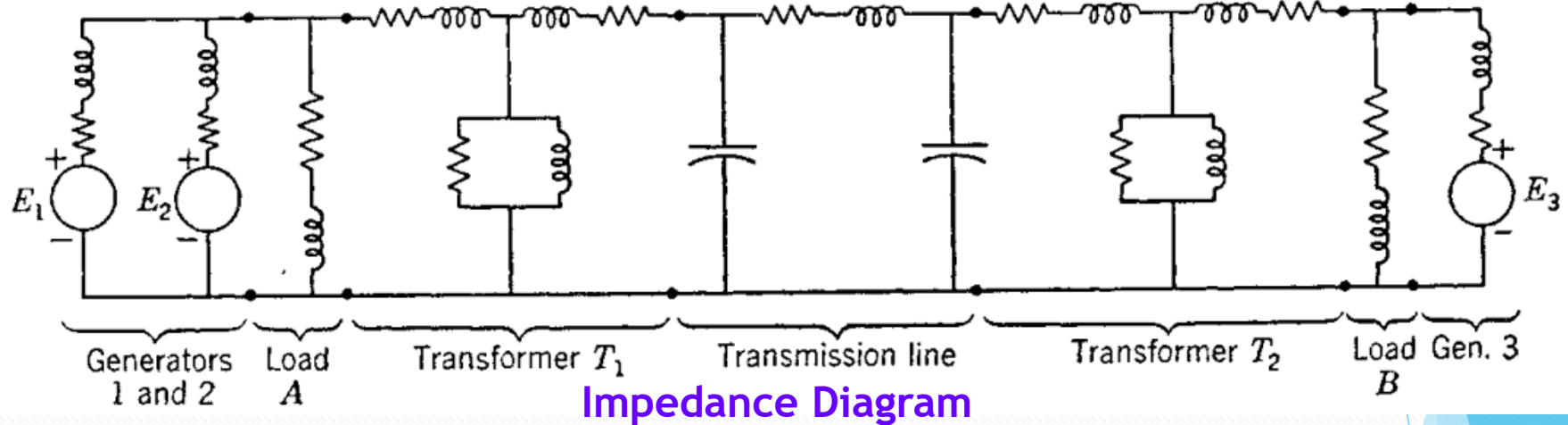
The impedance diagram can further be simplified by making certain assumptions and reduced to simplified reactance diagram.

Assumptions:

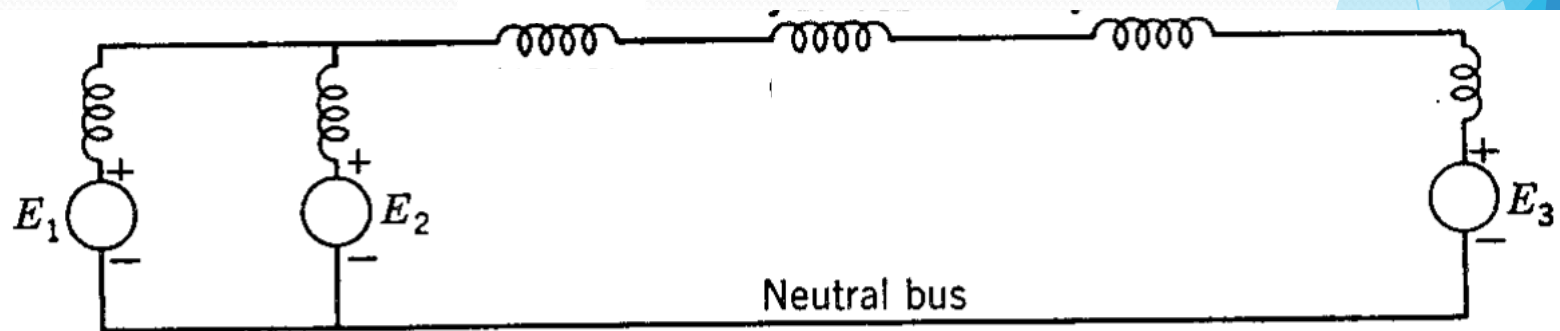
- ❑ Assumptions made for drawing reactance diagram give satisfactory results as inductive reactance of a system is much larger than its series resistance.
- ❑ Magnetizing current of a transformer is usually insignificant compared with the full-load current. So, shunt admittance is usually omitted in the equivalent circuit of the transformer.
- ❑ The capacitance of transmission lines are neglected.
- ❑ Loads which do not involve rotating machinery have little effect on the total line current during a fault and are usually omitted .
- ❑ The resistance is often omitted during the fault analysis. Therefore, resistance of generator armature, transformer winding resistance, transmission line resistance are neglected in reactance diagram.

Representation of Power System

IMPEDANCE DIAGRAM TO REACTANCE DIAGRAM

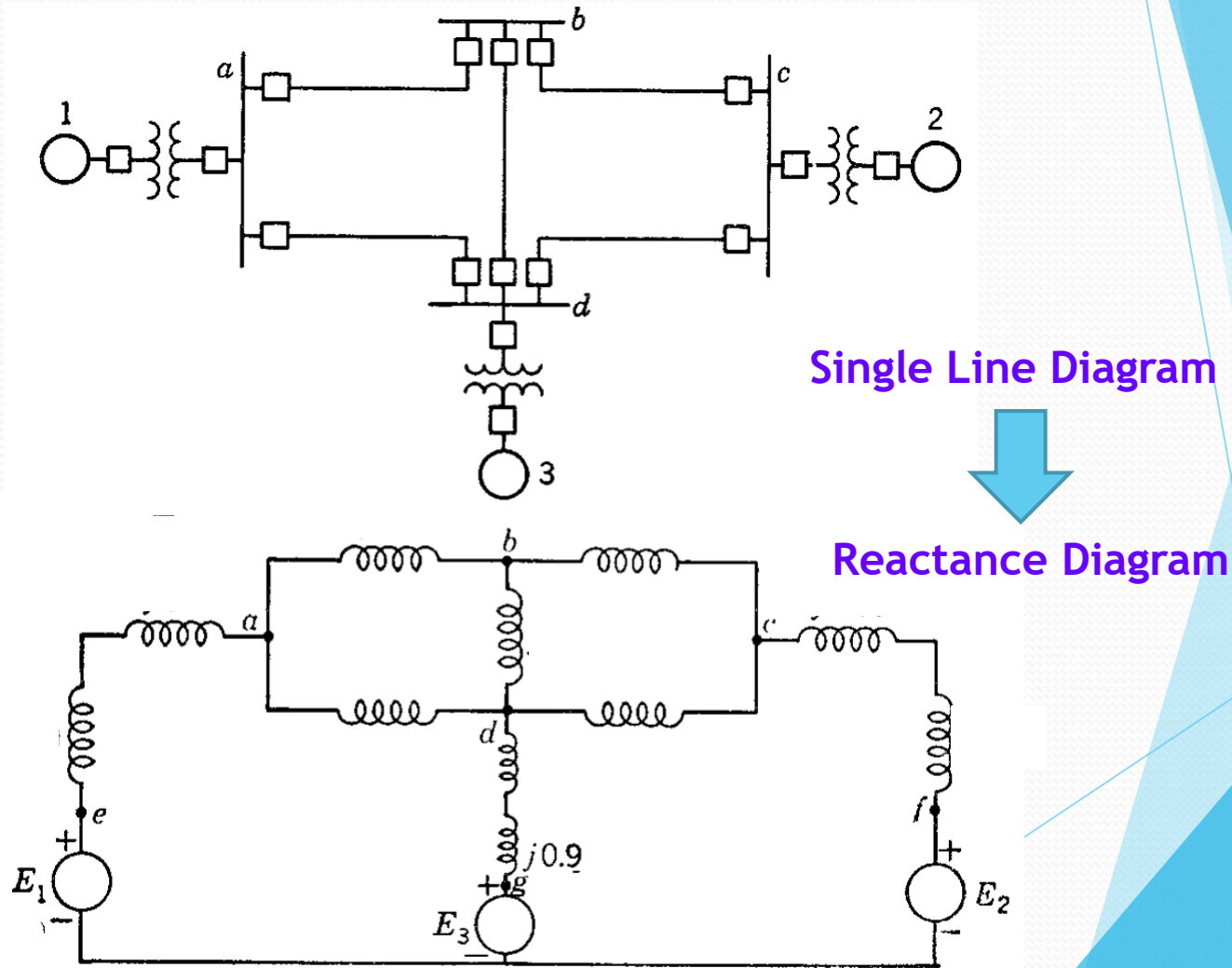


Reactance Diagram



Representation of Power System

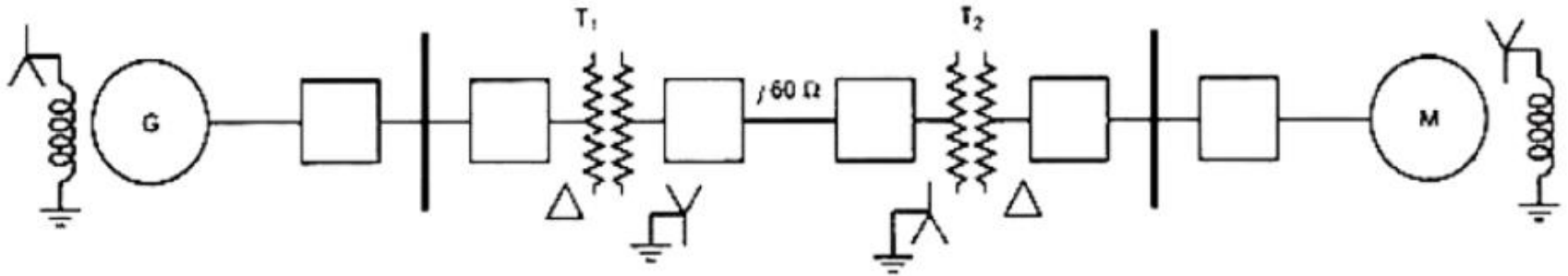
SINGLE LINE DIAGRAM TO REACTANCE DIAGRAM



Representation of Power System

IMPEDANCE DIAGRAM AND REACTANCE DIAGRAM

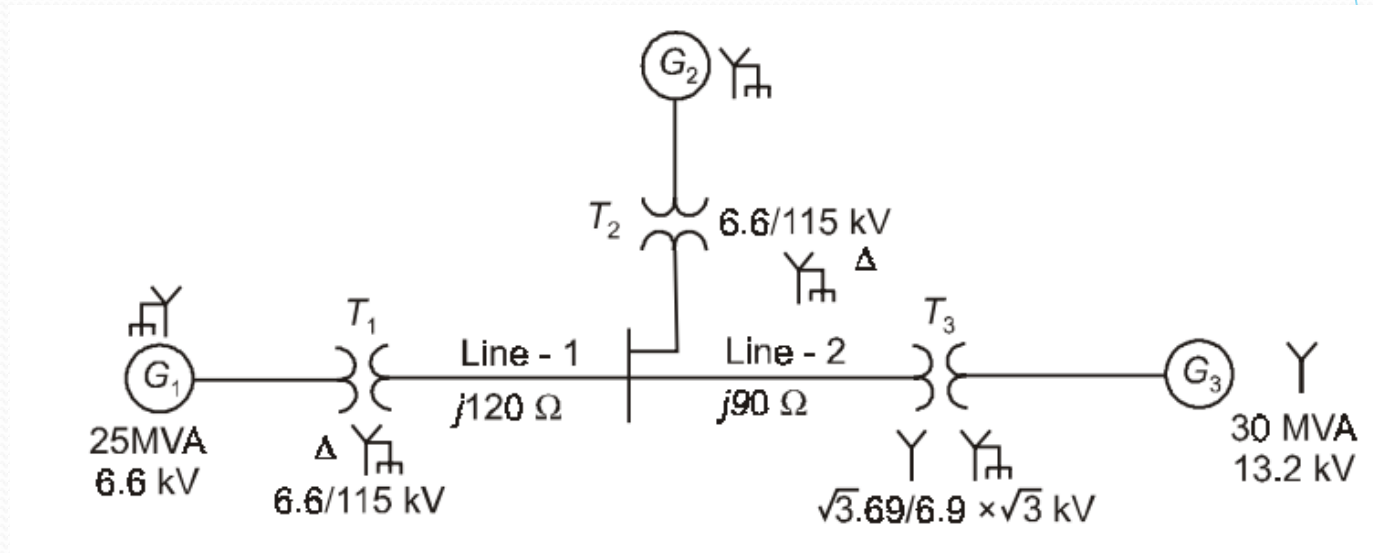
Practice Ques 1: Draw the impedance and reactance diagram of the following single line diagram of the power system.



Representation of Power System

IMPEDANCE DIAGRAM AND REACTANCE DIAGRAM

Practice Ques 2: Draw the reactance diagram of the following single line diagram of the power system.





QUESTIONS AND DISCUSSIONS

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